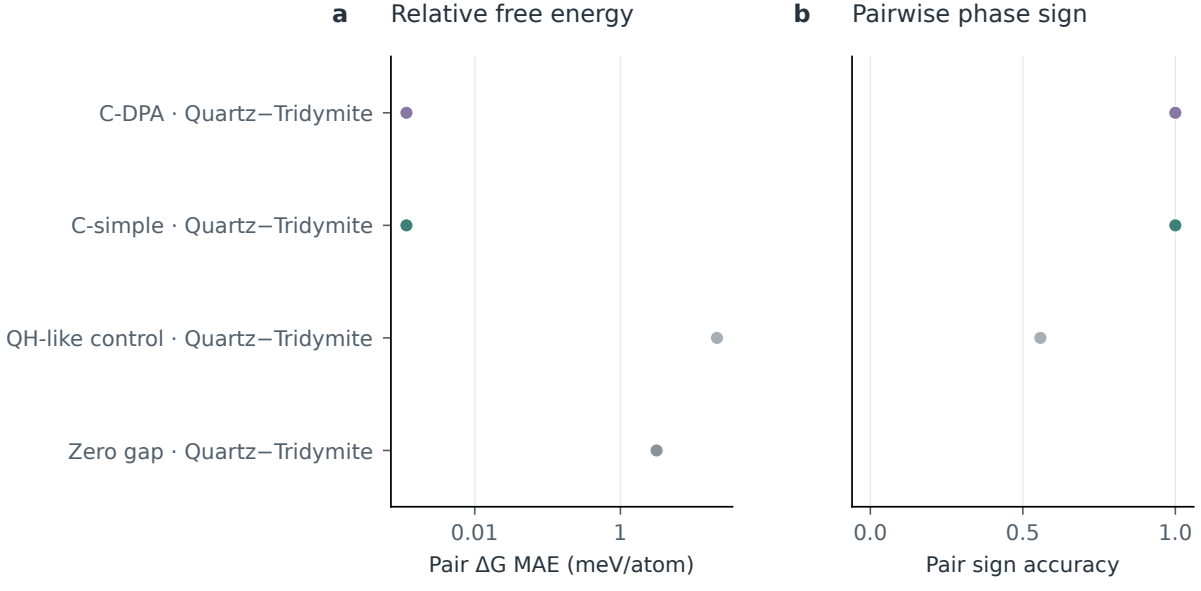
a
Relative free energy
C-DPA · Quartz−Tridymite
C-simple · Quartz−Tridymite
QH-like control · Quartz−Tridymite
Zero gap · Quartz−Tridymite
0.01
1
Pair ΔG MAE (meV/atom)
b
Pairwise phase sign
0.0
0.5
1.0
Pair sign accuracy